

WEEK 4 OVERALL LEARNINGS ON: Applied Cryptography and Common Attacks

Reflection:

During Week 4, I gained a deeper understanding of **applied cryptography** and the importance of implementing cryptographic systems securely. I learned that even strong algorithms like AES, ChaCha20, or RSA can become vulnerable if used incorrectly, highlighting the difference between theoretical security and real-world implementation. In particular, I explored **secure password storage** techniques such as salting and key stretching, including modern schemes like bcrypt, scrypt, and Argon2, and saw how weak hashes can be exploited through brute-force or dictionary attacks.

I also studied **randomness and entropy**, recognizing that cryptographic keys and nonces must come from secure random number generators, as predictable outputs can compromise entire systems. Through applied exercises with AEAD encryption, JWT tokens, and TLS concepts, I observed how misconfigurations or improper use can lead to **replay attacks, chosen-plaintext vulnerabilities, and integrity violations**.

This week reinforced that strong cryptography alone is not enough, and understanding the practical limits, secure design principles, and correct implementation is essential to building systems that are truly resilient against attacks. The hands-on exercises, including CTF challenges, helped me connect theory to practice and solidified my ability to identify and exploit insecure cryptographic practices in realistic scenarios.